# Week 40 Loop color certificates

Facilitator guide keyed to the revised student packets

## Mathematical purpose

A knot is a closed loop of cord that may be moved and bent but never cut, and whose strands may not pass through one another. Formally, it is a tame closed embedded loop in three-dimensional space, considered up to ambient deformation. A regular knot diagram records which strand goes over at each transverse crossing. A crossing in the picture is not a junction in the cord.

**Arcs and colorings.** An arc runs from one underpass to the next, continuing through overpasses. A crossing-free closed loop is one whole arc. Give every arc one of R, B, G. At each crossing the over-arc and the two under-arcs must be all one color or all three different. This guide counts all valid colorings, including constants. “Nontrivial” or “nonconstant” means more than one color is used.

**Exact target counts.** In the trefoil drawing, all three labeled arcs meet at every crossing. Its valid assignments are the three constant triples and all six permutations of R, B, G: nine total, six nonconstant. The plain loop has one arc and hence exactly three total colorings, all constant. The figure-eight drawing has four crossings and four arcs; only its three constant colorings are valid.

**Invariant and theorem.** Every deformation of a knot changes its diagram by a sequence of three local moves (Reidemeister’s theorem, supplied, not proved): a curl (Type I), the two-crossing move (Type II) and a slide of one strand across a crossing (Type III). Each move gives a one-to-one match between colorings before and after, with the same colors at the ends of the move and constant colorings matched to themselves. So the total number of colorings is the same for equivalent knots. Children check Type II with all nine end choices; grades 4–5 check the curl and the slide on the return page (27 inputs for the slide, shared out). Algebra for all three moves is on page 6.

**One-way certificate.** Different coloring counts prove knots inequivalent: the trefoil (9) is not the plain loop (3). Equal counts do not prove equivalence: the figure-eight knot, drawn in grades 4–5 Problem 7, has only the three constant colorings, like the plain loop, yet it is knotted (with five colors it has 25 colorings and the plain loop 5). Failing to untangle a cord is not an impossibility proof; a successful legal untangling is evidence of equivalence for that actual model.

## Band mapping and gates

**All bands** start by counting the figure-eight’s whole arcs (Problem 1, four), which checks whether a child follows an arc through its overpasses. **K–1:** complete single crossings (Problem 2), then find the trefoil’s colorings and compare the plain loop and the cords. **Grades 2–3:** exhaust the trefoil colorings and establish unique completion in the two-crossing move. **Grades 4–5:** also check the curl and the slide, use the supplied theorem to show the trefoil is knotted, and meet its limit in the figure-eight. Experiments (the cords) are kept apart from certificates (the counts). No modular arithmetic is required of children.

**Before-use gate:** exact cord models, print/marker channel fit, and child handling have not been physically pretested. A child who counts eight pieces in Problem 1 is counting segments between crossings; follow one arc on the physical overpass before any coloring. Keep all cords on the table; children never cut or wear loops. Diagram checks do not certify the physical models.

## Preparation, demonstration and a 40-minute route

**Materials.** Per table: four short, thick, flexible closed cords, about 400–600 mm each, in a tray, tagged A to D with masking tape: A a plain loop, B a complicated unknot, C a trefoil, D a complicated unknot (keep this key with the adults). Per child: the packet, tracing paper, and R/B/G markers paired with letters or distinguishable patterns. The trefoil and the grades 4–5 figure-eight are 148 mm wide; do not shrink them. The 2.4 mm coloring channel and underpass gaps still need a test with your actual markers.

**Adult model preparation and required physical test.** Close each cord by overlapping its ends about 3 cm and taping them flat; never tie the ends, which puts a knot into every loop. Make the complicated unknots by reversible motions from a plain loop (add a curl, lift a bent strand over another segment) or by closing a slipped overhand knot; record and reverse your sequence. For the trefoil, tie a loose overhand knot, tape the ends, and lay it out to match the printed three-crossing picture. An overhand knot tied the other way round is its mirror image and cannot be laid out to match; if every crossing comes out reversed, retie it. Predictions for the pretest: a curl in a plain loop falls open when the loop is lifted; a taped slipped overhand knot is an unknot that keeps its knotted look until it is worked loose. Verify each whole cord, every over/under relation and the join. Reject damaged cords, exposed wire, tight knots, electrical cords and unsafe small closures.

**Launch, about 5 minutes.** Let children handle the tray briefly. On student page 1, start just after the left underpass, follow the horizontal strand over the middle crossing, and stop just before the right underpass; do not switch to the vertical strand. Point to the continuous overstrand in the three crossing examples: R/R/R and R/B/G are valid; R/R/B is not. Then color one crossing wrongly on the board (R over, R and B under) and let children catch it. **Checking crossings:** the rule lives only on paper, so before a record is filled in, a partner (in grades 4–5, a child acting as referee) points to the over-arc and both under-arcs at every crossing and says “all same” or “all different”.

| Band | First visit: launch, then about 35 minutes at the tables | Return visit |
|---|---|---|
| K–1 | Problems 1–5: count arcs, complete crossings, color the trefoil and the plain loop, try cords A–D. If time runs short, try the cords before finishing the trefoil list. | Problem 6: the two crossings and six end-color cases. |
| Grades 2–3 | Problems 1–5: count arcs, all trefoil colorings, the plain loop, cords A–D, and the nine-row table for the two-crossing move. | Problem 6: why each choice has one completion. |
| Grades 4–5 | Problems 1–5: as grades 2–3, with the explanation in Problem 5. | Problems 6–7: the curl and the slide, the certificate, and the figure-eight. |

**Readiness.** If a K–1 child’s tracing switches strands in Problem 1, use the physical overpass again before any coloring. Adults prepare and cut materials.

## K–1 answer key

**Problem 1.** Four. Each of the four underpasses ends one arc and starts the next. Eight is the count of pieces between all crossings; follow one arc through its overpass to see that the pieces join.

**Problem 2.** In the printed 3-by-3 arrangement, the left under-color is fixed by the row (R, B, G) and the over-color by the column (R, B, G). Missing right under-colors are:

| Left under-color | Over R | Over B | Over G |
|---|---|---|---|
| R | R | G | B |
| B | G | B | R |
| G | B | R | G |

No question mark has two answers. If the known colors agree, the third must agree. If they differ, the third must be the remaining color.

**Problem 3.** Reading a record as (arc 1, arc 2, arc 3), the complete set is RRR, BBB, GGG, RBG, RGB, BRG, BGR, GRB, GBR. There are nine, not twelve; the extra recording places do not announce the answer. Exactly two equal arc colors would put “two same and one different” at every crossing and are invalid.

**Problem 4.** No. The plain loop has only one whole arc, so its entire length must have one color. Its three possible colorings are all red, all blue, or all green.

**Problem 5.** For correctly prepared models, A, B and D can become the plain shape; C, the trefoil, cannot. Record what actually happens. A stuck cord does not prove impossibility; an unexpectedly untangled C may have been assembled incorrectly. No global proof is expected of this band.

**Problem 6.** Cases are numbered on the page. L, P, S, Q are end-dot colors, not new names for separate arcs.

| Case | (L,P,S,Q) | Works | Middle horizontal color |
|---|---|---|---|
| 1 | (R,B,R,B) | Yes | G |
| 2 | (R,R,B,R) | No | The second crossing forces S to be R, not B |
| 3 | (G,B,G,B) | Yes | R |
| 4 | (B,G,B,R) | No | P and Q disagree on one whole arc |
| 5 | (B,B,B,B) | Yes | B |
| 6 | (R,G,R,G) | Yes | B |

Successful cases have P=Q and L=S. The unique middle color follows the Problem 2 table; passing under the same color twice restores the starting color.

## Grades 2–3 answer key and target verification

**Problem 1.** Four whole arcs, one for each underpass. A child who counts eight pieces has stopped at overpasses too.

**Problem 2.** Arc-order triples: RRR, BBB, GGG, RBG, RGB, BRG, BGR, GRB, GBR. All nine satisfy all three crossings. To show completeness, pick colors for arcs 1 and 2. If they agree, arc 3 is forced to agree; if they differ, arc 3 is forced to be the remaining color. There are 3×3=9 choices of the first two colors, each with exactly one completion.

| Crossing in final drawing | Over-arc | Two under-arcs |
|---|---|---|
| Upper left | 3 | 1 and 2 |
| Upper right | 2 | 3 and 1 |
| Lower center | 1 | 2 and 3 |

This is the actual final crossing/arc relation, checked against the generated spatial trefoil and visible gaps. It matters more than the silhouette: reversing one crossing can change the knot. The three printed numbers label whole arcs, not regions enclosed by the curve.

**Problem 3.** Exactly three colorings, one constant choice each of R, B, G; none uses more than one color because the crossing-free loop is a single arc.

**Problem 4.** Expected for correctly assembled models: A, B and D can be legally opened into the plain shape; C, the trefoil, cannot. After the attempts, ask: “What can your attempts tell you, and what remains uncertain?” Successful legal motions establish what was achieved; failure within a short session leaves the question unresolved by experiment. A formal trefoil obstruction uses the invariant, not frustration or elapsed time.

**Problem 5.** All nine ordered choices of (L,P) work. In the before picture S=L and Q=P; the middle color is L if L=P and the third color otherwise. Rows (L, P, middle, S, Q): RRRRR, RBGRB, RGBRG, BRGBR, BBBBB, BGRBG, GRBGR, GBRGB, GGGGG. In the after picture the whole horizontal strand is L and the bent strand is P, so the end dots match.

**Problem 6.** No. For fixed L and P, the first crossing has only one possible middle color, and the second crossing, under the same bent strand, has only one possible S, which is L again. The bent strand is one arc, so Q=P. The after picture is also colored in only one way by its two strands. This gives a one-to-one match keeping all four end-dot colors, not just a comparison of how many examples happened to work.

## Grades 4–5 answer key

**Problem 1.** Four whole arcs. Each underpass ends one arc; a count of eight stops at overpasses too.

**Problem 2.** Nine total colorings: RRR, BBB, GGG and the six permutations RBG, RGB, BRG, BGR, GRB, GBR in labeled arc order (1,2,3). Three are constant and six nonconstant. Completeness follows from the first-two-arcs argument on page 4; every crossing involves all three arcs, so a choice valid at one is valid at the others.

**Problem 3.** The plain loop has exactly three colorings, since it has exactly one arc. Thus the totals differ: 9 versus 3. Do not count rotations of the paper or permute the arc labels as an equivalence relation; the worksheet uses labeled arcs and fixed named colors.

**Problem 4.** State observations separately from conjectures. A, B and D, correctly built, can be moved to the plain shape. Not managing C in class is only a failed attempt. The count argument in Problem 6 supplies the impossibility certificate for the trefoil.

**Problem 5.** Every ordered pair (L,P) from R/B/G works, and each has one completion in each picture. Write L=a, P=b, and identify colors with 0, 1, 2 modulo 3. At the first crossing the middle under-color is $c=2b-a$. At the second it becomes $2b-c=a$. The continuous bent over-arc gives $Q=b$; hence $S=a$. In the after picture those are the two whole-strand colors. The nine-row table is the same operation without modular notation.

**Problem 6.** *Curl:* the crossing has the loop's arc over and both J's arc and the loop's arc under, so J's arc must match: the whole curl is one color, J=K, as after. *Slide:* with T, U, V chosen (27 ways, shared out), U–Y never passes under, so Y=U; before and after, $Z=2U-T$ and $X=2T-2U+V$ (mod 3), so the end colors match and each choice has one completion. *Certificate:* No. Every deformation is made of the three moves (supplied), each keeps the count, and the trefoil has 9 colorings against the plain loop's 3. Children have checked the moves, not proved the supplied theorem.

**Problem 7.** Only RRRR, BBBB, GGGG. Crossings (over; under, under): (4; 1, 2), (3; 4, 1), (1; 2, 3), (2; 3, 4). If arcs 1 and 2 differ, arc 4 is the third color, then arc 3 equals arc 2, and the crossing where arc 1 is over has two same and one different. Equal counts only mean this invariant has not distinguished two loops; the figure-eight and the plain loop both have 3 and differ. Neither "three colorings" nor "no colorings with more than one color" proves that a knot is an unknot.

## Optional hints

For exhaustive coloring: "Choose the colors of two labeled arcs; what is left for the third?" For the local move: "Does the bent strand ever pass under anything?" For the certificate: "If a legal deformation existed, what number would have to stay the same?" Never change an over/under crossing just to make the coloring fit. For the slide: "Which strand never goes under?"

## The full invariant and before-use record

## Adult proof of the local algebra

Encode colors in the field with three elements. With incoming under-color a and over-color b, the unique outgoing under-color is a*b=2b−a modulo 3. This is equivalent to the printed all-same/all-different rule. The operation satisfies the three Reidemeister identities:

**Type I:** a*a=a.
**Type II:** (a*b)*b=2b−(2b−a)=a.
**Type III:** (a*b)*c=a−2b+2c, while (a*c)*(b*c)=2(2c−b)−(2c−a)=a−2b+2c.

These identities and the unique crossing completion give invertible local extensions with unchanged colors outside the move disk. The inverse move recovers the original coloring, so the correspondence is a bijection. Constant colorings correspond to themselves. Reidemeister's theorem then supplies the passage from local moves (with planar isotopy) to all equivalences of tame knots. Therefore total and nonconstant coloring counts are invariants. This argument does not make matching counts a complete test.

## Primary sources and precise use

Mladen Bestvina, *Knots: a handout for Summer Math Program* (2007), §§1-4, especially §2 on Reidemeister moves and §3 on tricolorability. For the figure-eight counterexample, §4, Problem 11 (PDF p. 8) gives n-colorability exactly when 5 divides n, ruling out nonconstant 3-colorings and supplying a nontrivial 5-coloring obstruction to the unknot. Handout PDF.

Louis H. Kauffman, *Knots*, "The Three Colored Trefoil" and "The Quandle and the Determinant of a Knot," gives the coloring obstruction and the operation 2b−a. The target counting and endpoint-case keys here are independently enumerated from the final diagram. Author's exposition.

The printed trefoil is an original rendering of (sin t+2 sin 2t, cos t−2 cos 2t, −sin 3t), with depth deciding over/under; underpasses delimit its three arcs. The figure-eight is the plane curve (cos t−1.5 cos 3t+0.5 sin 2t, sin t+0.5 sin 2t−1.5 sin 3t), its four crossings taken alternately over and under: 3 three-colorings, 25 five-colorings. Numerical and visual checks establish the drawings; they do not construct or certify the classroom cord.

## Record the required physical checks

Date and adult: _____
Cords A–D taped and checked (crossings, trefoil handedness): _____
Both complicated unknots (B, D) successfully undone by recorded legal moves: _____
100% print, marker channel, tracing paper, and cord handling tested: _____
K–1 arc counting in Problem 1 observed: _____

**Status:** exact-material checks above remain unperformed here. This guide and the revised packets are unpiloted. Keep loops on the table, allow no child cutting, and make no claim that a rendered diagram certifies a successful physical model.

Key alignment: F40-K-v3 has 5 pages and Problems 1–6; F40-23-v3 has 4 pages and Problems 1–6; F40-45-v3 has 6 pages and Problems 1–7. Prepared October 3, 2026; revised October 23, 2026 from the review card.

# Starting routes and return visits

Planning update of 4 October 2026, revised 23 October 2026. The timing menus are examples; a theme can continue over several meetings. Keep the full Grades 4-5 investigations and choose by readiness.

## Use and return choices

Keep one adult at each table to manage cords A–D and to referee crossings. K–1: Problems 1–5 on a first visit; if counting arcs in Problem 1 shows a child switching strands, stay with the physical overpass before coloring. Problem 6 (two crossings, six end-color cases) is for a return visit. Grades 2–3: Problems 1–5 first; Problem 6 on return. Grades 4–5: Problems 1–5 first; Problems 6–7 (the curl and the slide, then the figure-eight) on return. Actual cord/marker fit and model pretests remain unperformed.